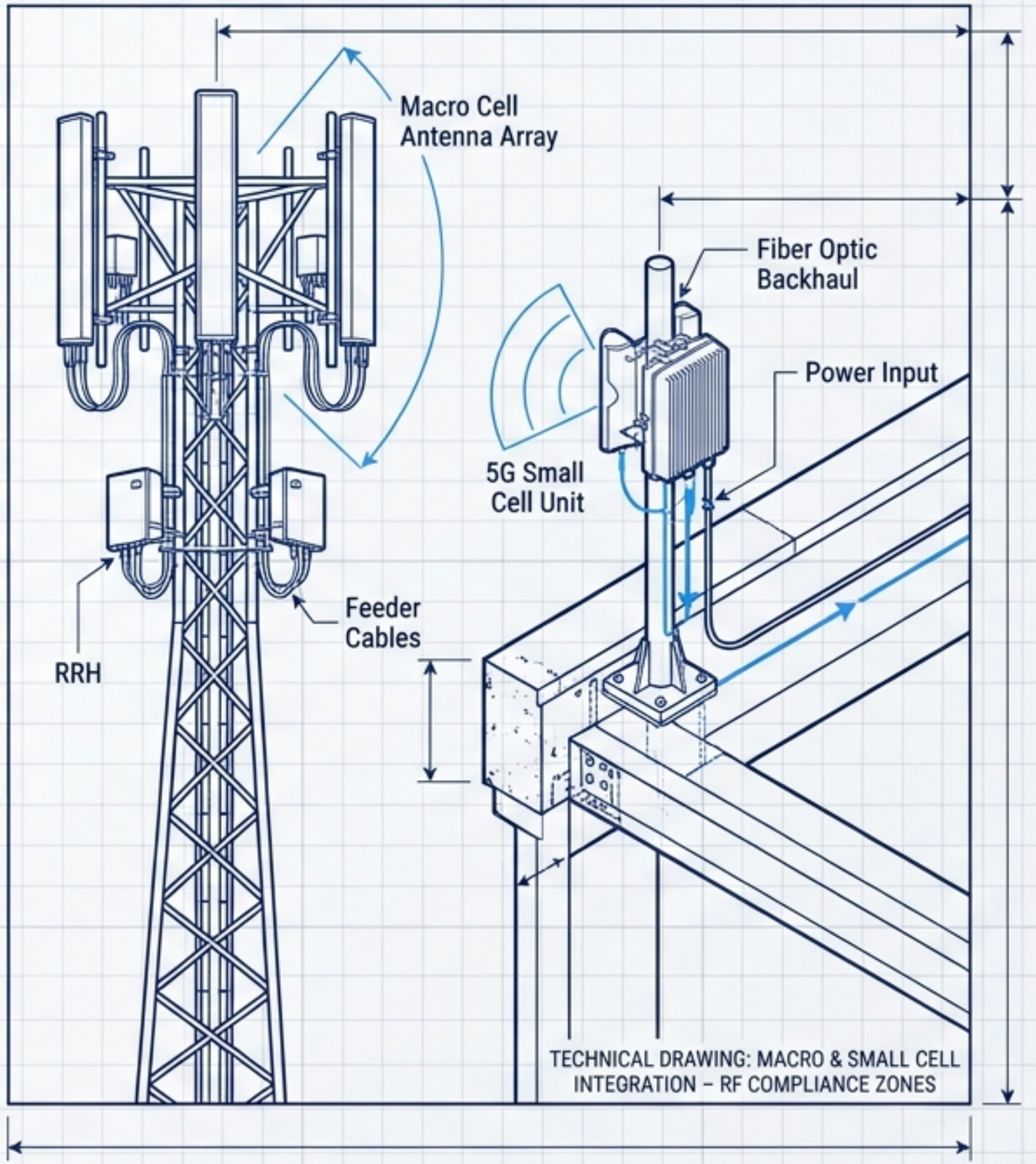
SC6 RF Exposure Compliance

The definitive assessment process for wireless infrastructure.

Scope Parameters:

Standard:	Health Canada Safety Code 6
Frequency Range:	3 kHz to 300 GHz
Applications:	4G/5G, Rooftops, Macro Sites, Small Cells



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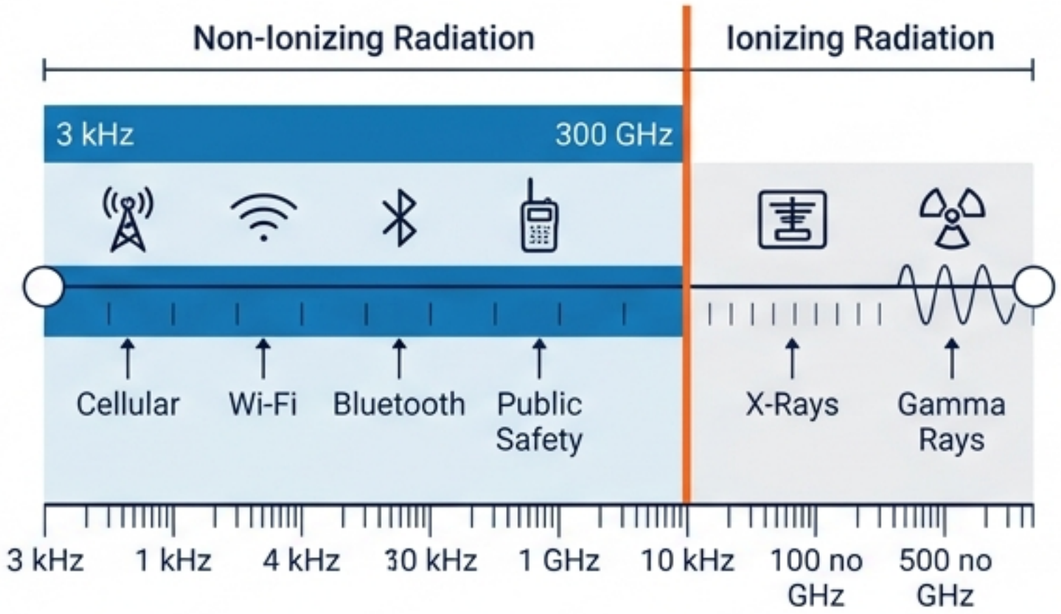
Demystifying SC6 and RF Energy

THE STANDARD

Safety Code 6 (SC6) is Health Canada's standard for human exposure to radiofrequency (RF) electromagnetic fields.

> **The '6' in Safety Code 6 is simply a document number in a Health Canada series. It does not represent a frequency, power level, or technical parameter.**

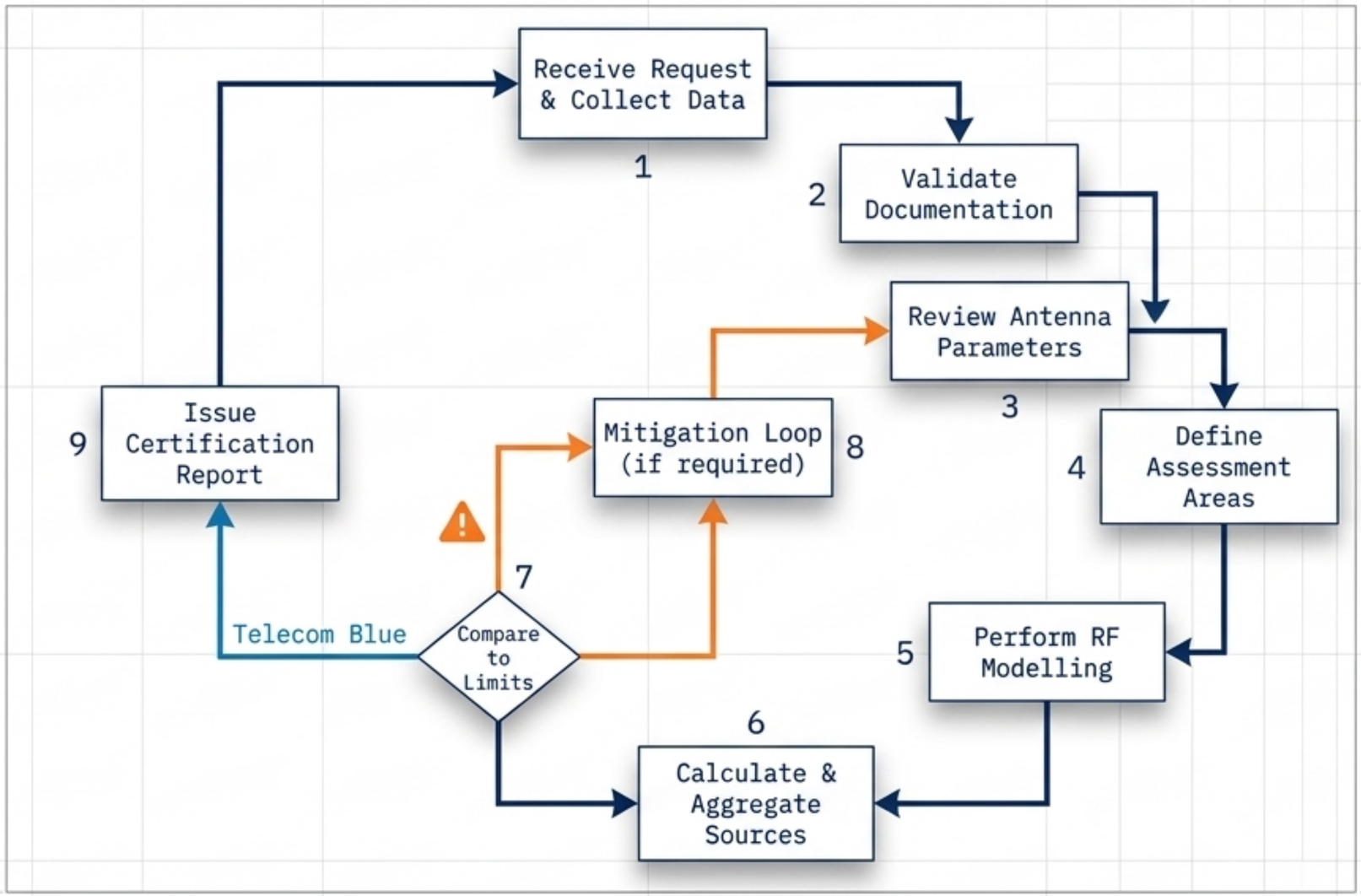
THE SPECTRUM



Key Takeaway: RF energy consists of electric and magnetic fields that do not ionize atoms.

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The Master Assessment Workflow



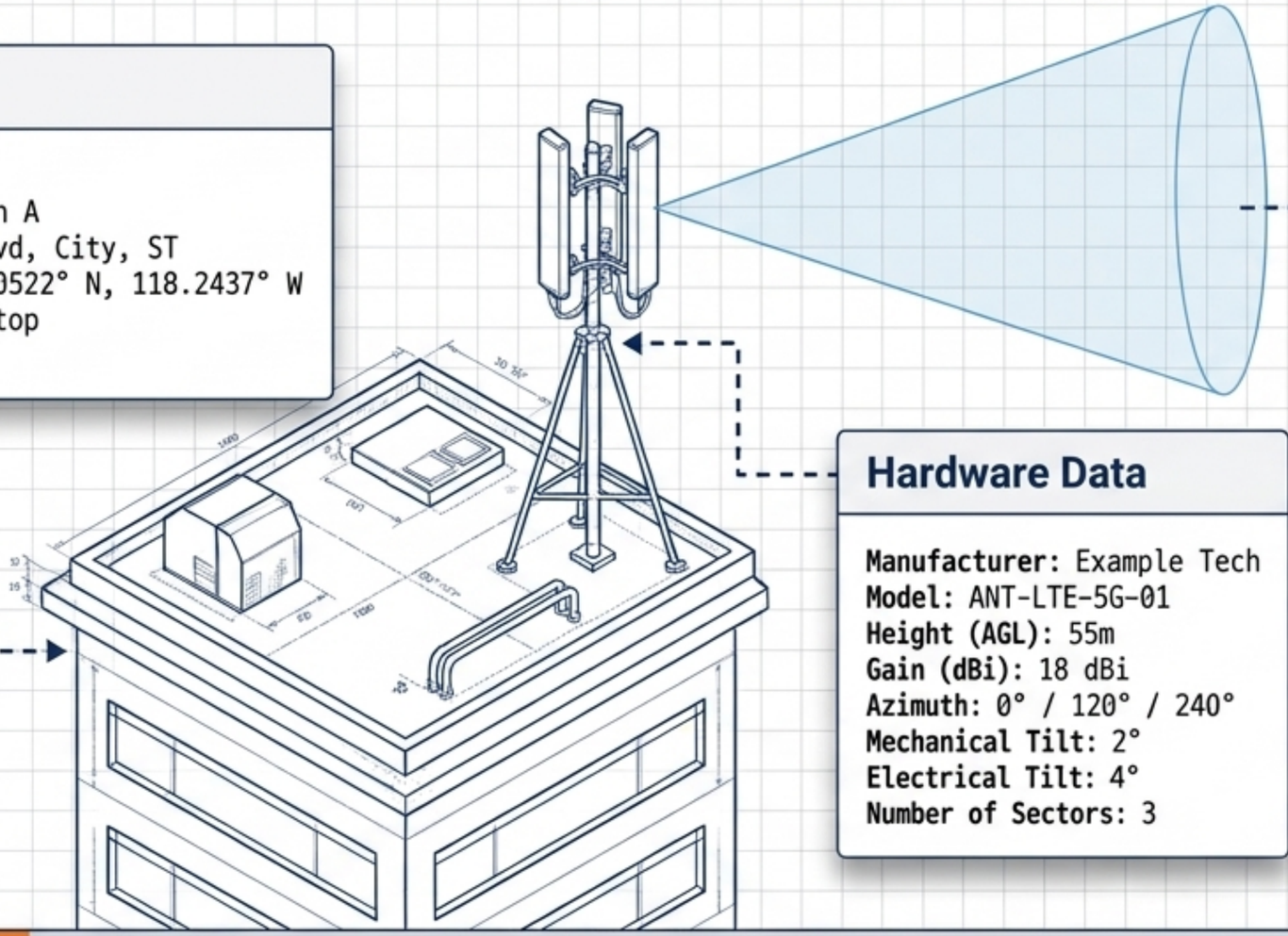
Objective: Verify that radiofrequency (RF) exposure levels remain below the limits established by Health Canada.

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Phase 1: Harvesting the Site Anatomy

Site Data

ID: SITE-001
Name: Rooftop Station A
Address: 123 Tech Blvd, City, ST
GPS Coordinates: 34.0522° N, 118.2437° W
Classification: Rooftop
Elevation: 50m AGL



Transmission Data

Freq Bands: 700, 1900, 2100, 3500 MHz
LTE/5G NR Carriers: LTE: 2x20MHz, 5G NR: 1x100NHz

Transmit Power: 4 x 40W / sector
EIRP: 58 dBm
MIMO: 4x4 MIMO
Carrier Aggregation: 3CA + NR

Hardware Data

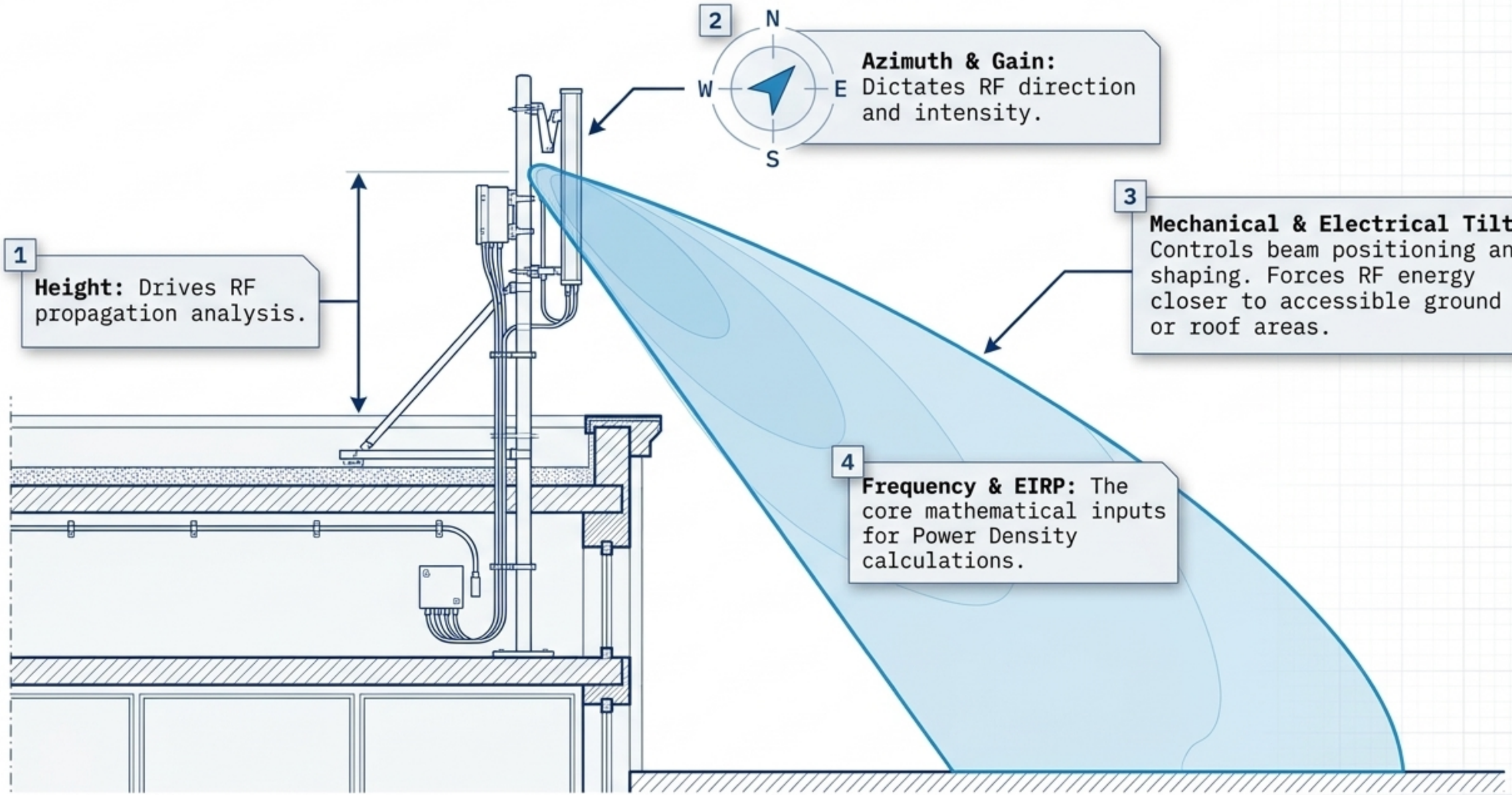
Manufacturer: Example Tech
Model: ANT-LTE-5G-01
Height (AGL): 55m
Gain (dBi): 18 dBi
Azimuth: 0° / 120° / 240°
Mechanical Tilt: 2°
Electrical Tilt: 4°
Number of Sectors: 3

✓ Pass/Fail

Before modelling, the analyst must confirm site drawings, access areas, antenna models, and power levels are documented. Missing data triggers a mandatory information request.

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Phase 2: How Antenna Parameters Shape Exposure



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Metric Alignment: Power Density vs. Specific Absorption Rate (SAR)

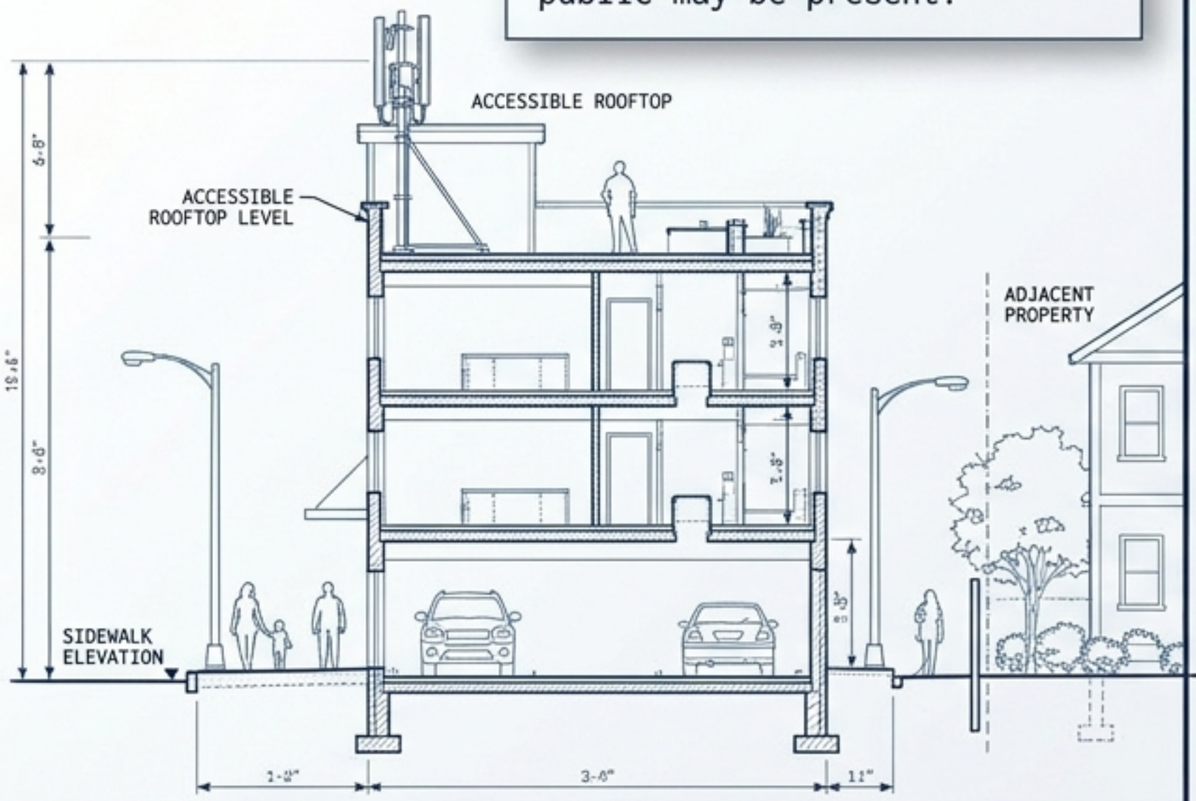
Power Density	SAR
	
Definition: Measures RF energy present in an area (Outside the body).	**Definition:** Measures RF energy absorbed by human tissue (Inside the body).
Unit: W/m^2 (Watts per square meter).	**Unit:** W/kg (Watts per kilogram).
Application: Used for Cell Towers, Rooftop Sites, Small Cells, Broadcast Systems. (Standard for SC6 assessments).	**Application:** Used for Mobile Phones, Tablets, Wearables, devices operating close to the body.

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Phase 3: Defining the Assessment Environments

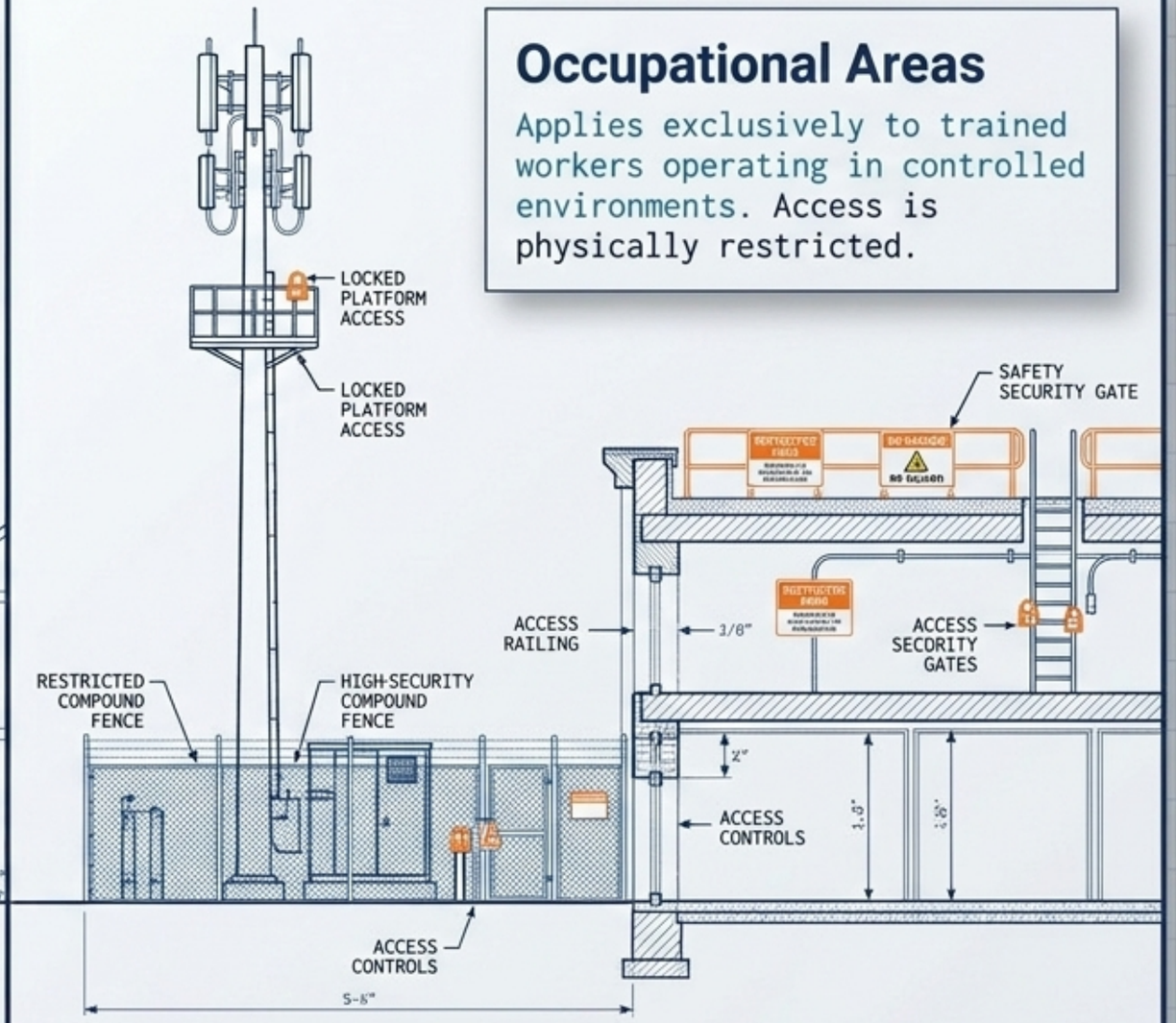
Public Areas

More restrictive limits apply. Exposure calculations are required anywhere the general public may be present.



Occupational Areas

Applies exclusively to trained workers operating in controlled environments. Access is physically restricted.



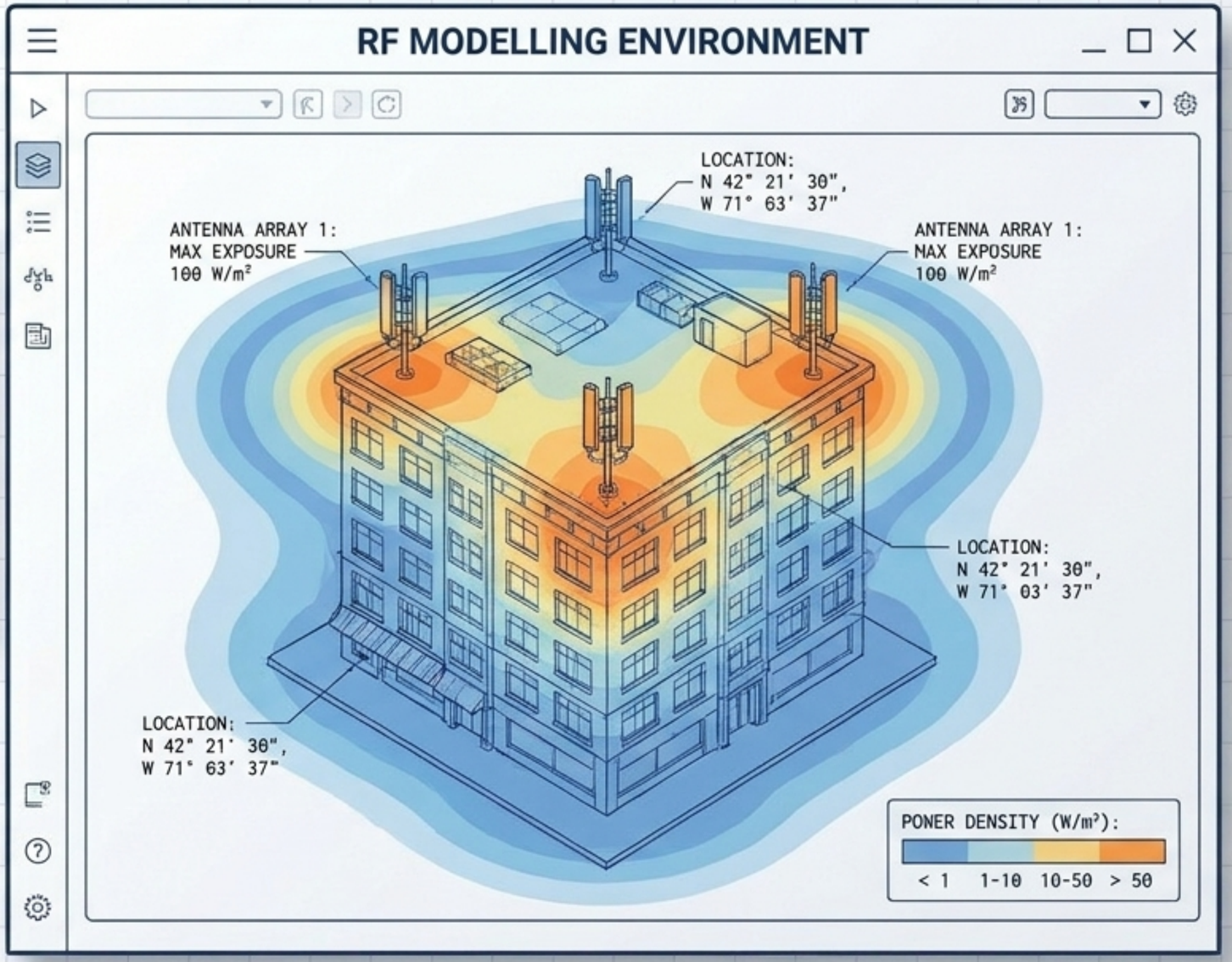
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Phase 4: RF Modelling and Output Generation

TOOL STACK

Standard Modelling Engines:

- EMF Visual
- Atoll
- EDX SignalPro
- Proprietary Tools



OUTPUT

Primary Phase Deliverables:

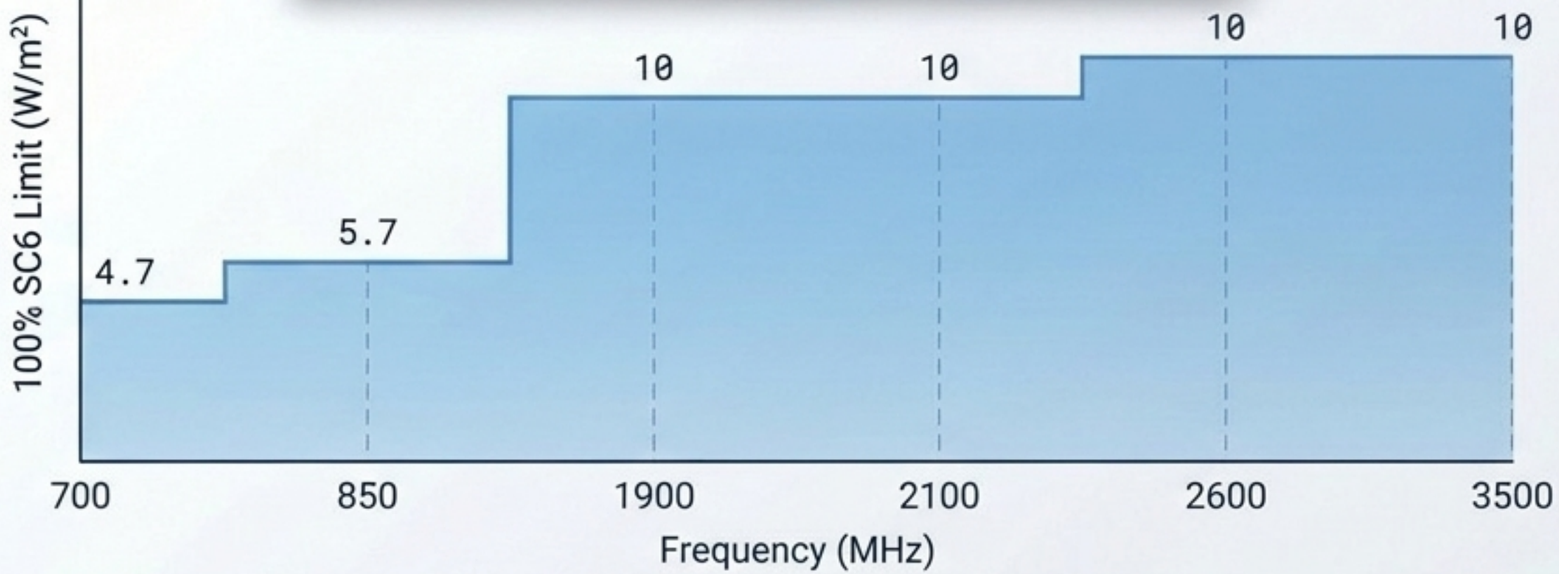
1. Power Density metrics
2. Visual Exposure Contours
3. Spatial Compliance Boundaries
4. Exposure Percentages

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Phase 5: SC6 Exposure Limits Scale with Frequency

Why does the limit change?

RF energy interacts differently with the human body at different frequencies. Tissue absorption dictates the safety threshold.



Math Widget

Formula:

$$\% \text{ SC6} = \frac{(\text{Calculated Power Density}}{\text{SC6 Limit}) \times 100} \times 100$$

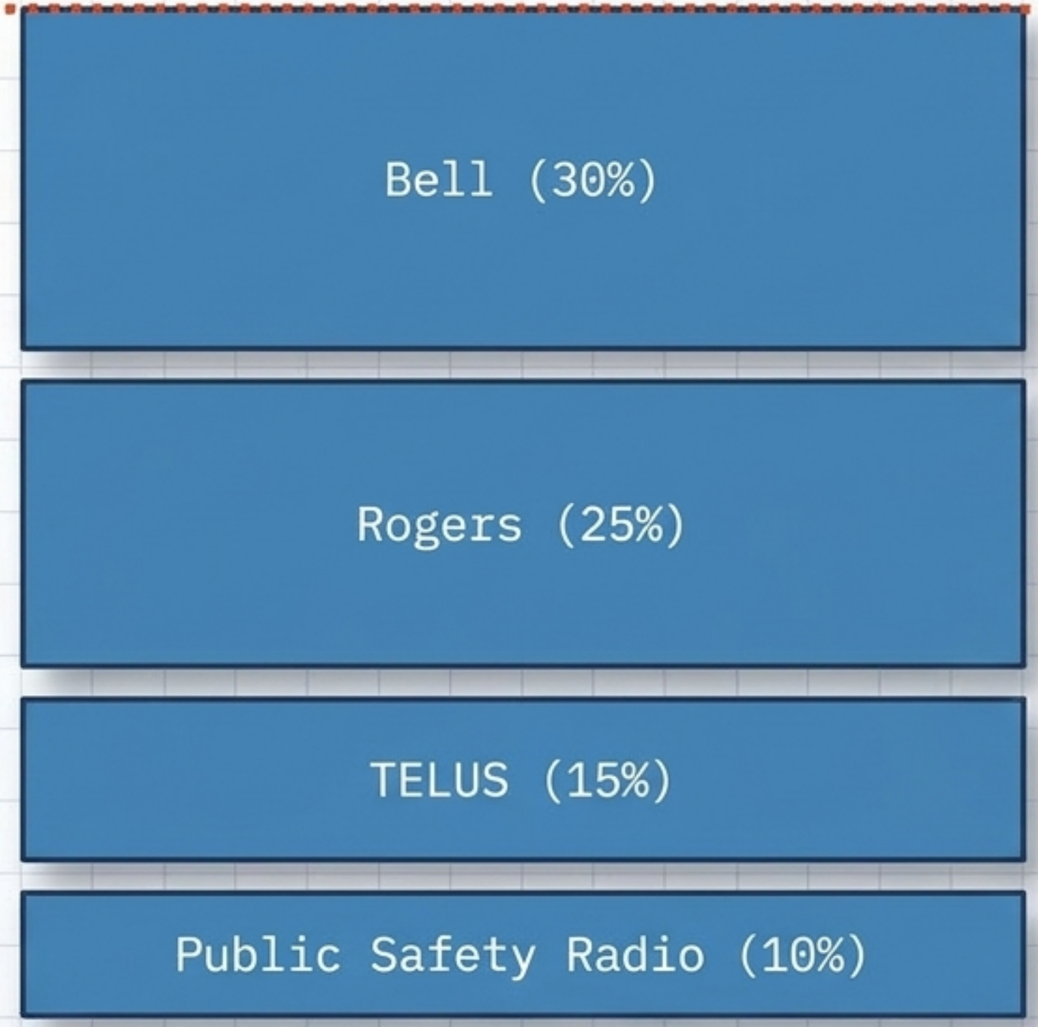
Example:

At 2100 MHz, the limit is 10 W/m². A calculation of 7.5 W/m² equals 75% SC6. Because 75% is less than 100%, the location is compliant.

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Phase 6: The Cumulative Impact of Multiple Sources

100% SC6 Regulatory Limit



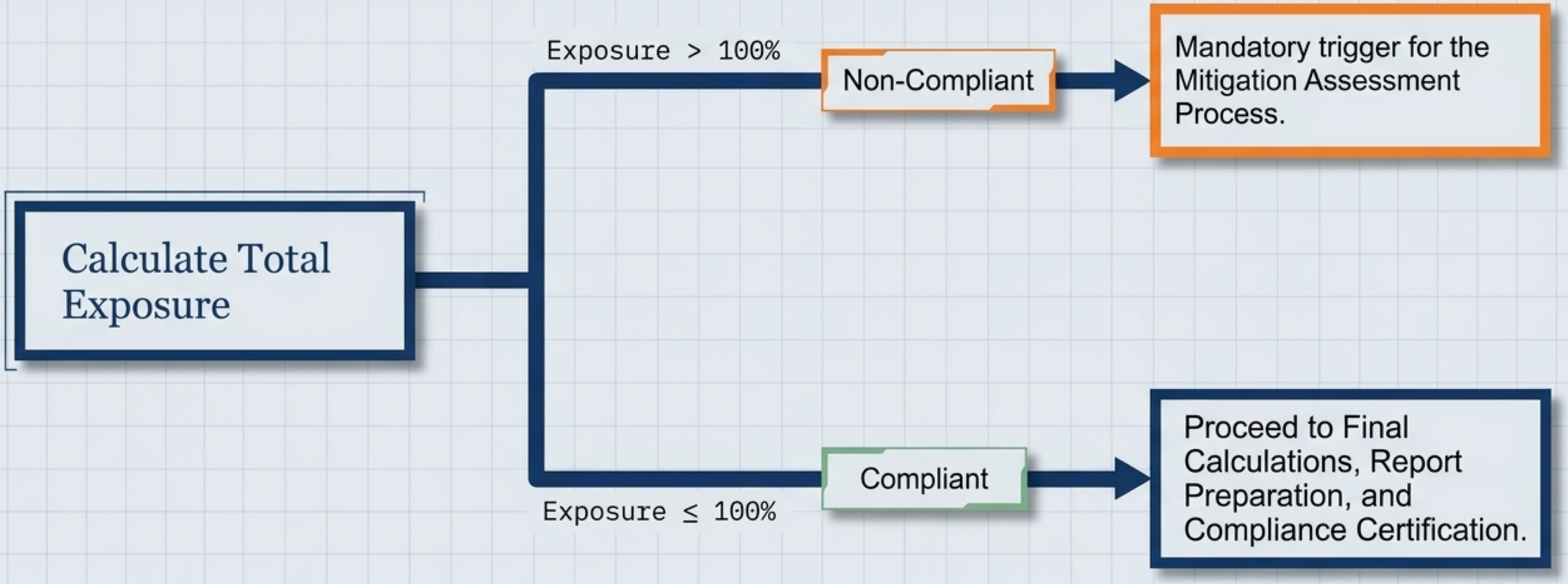
The Aggregation Stack

Total Exposure: 80%

Where multiple RF systems exist (including broadcast and microwave links), all exposure sources must be combined. The total aggregate percentage is evaluated against the 100% ceiling.

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The Compliance Decision Point



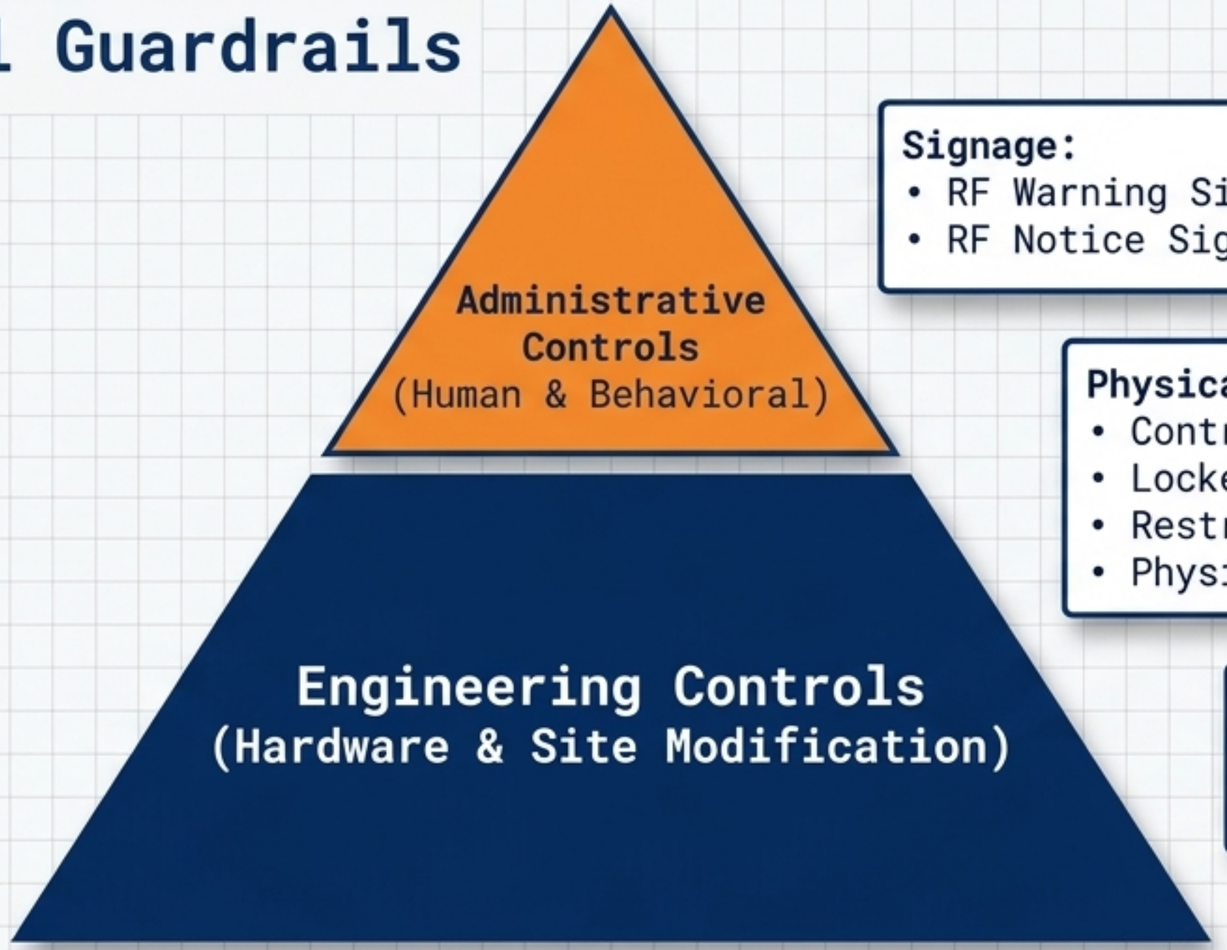
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Engineering Controls: The Mitigation Diagnostic Matrix

Cause	Example	Engineering Fix
High Transmit Power	LTE Additions / 5G Overlays	Reduce Power, Traffic Balancing, Network Redesign
Antenna Orientation	Excessive Main Beam Exposure	Adjust Azimuth, Modify Mechanical/Electrical Tilt
Insufficient Separation	Walkway in RF Beam	Increase Setback, Relocate Access Route
Antenna Placement	Mounted Too Low	Increase Mounting Height, Change Structure
Multi-Carrier Conflict	Shared Rooftop	Coordinate with Co-Located Operators, Reassess Combined EIRP

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Administrative Controls and Behavioral Guardrails



Signage:
- RF Warning Signs
- RF Notice Signs

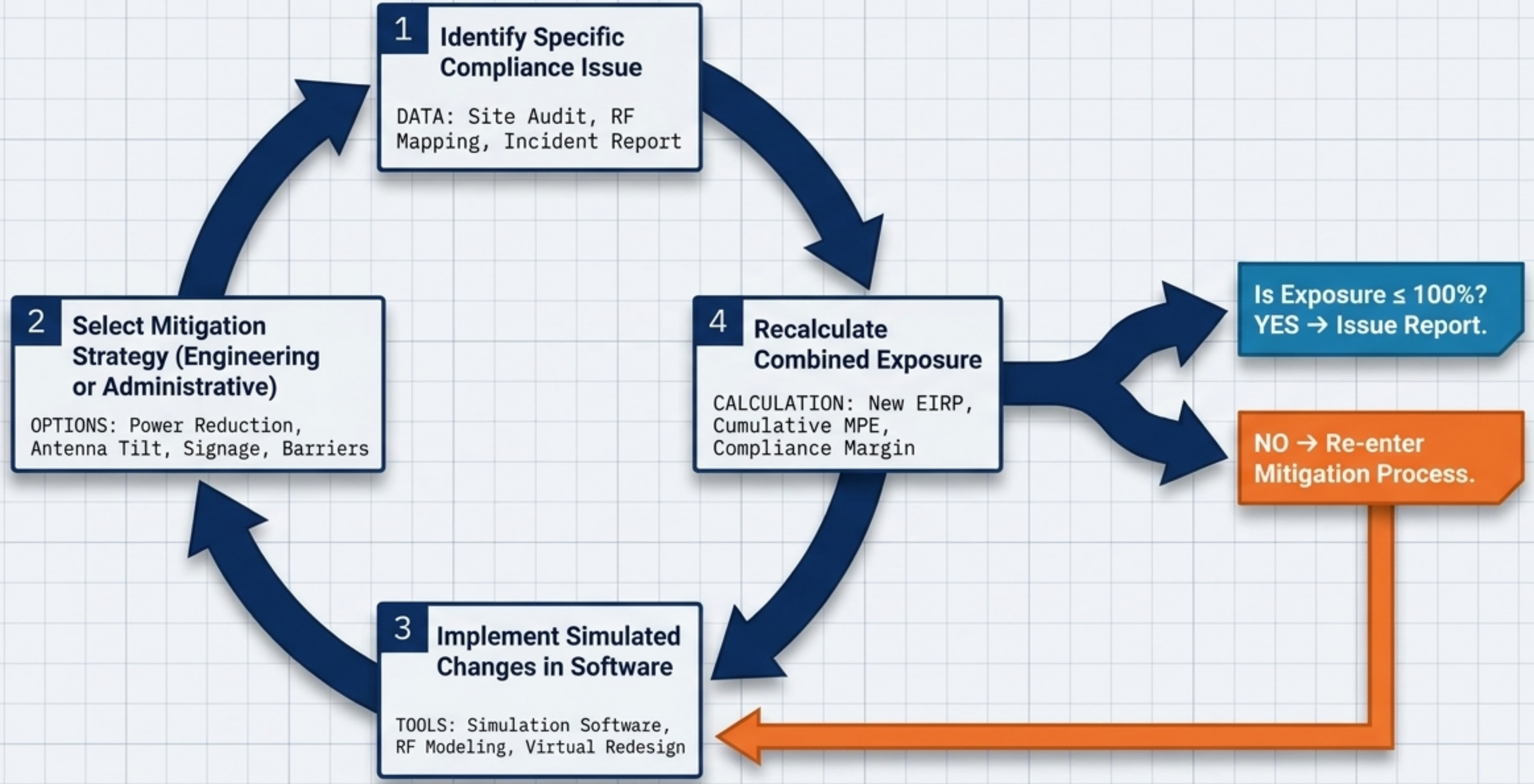
Physical Deterrents:
- Controlled Access Zones
- Locked Gates
- Restricted Rooftop Access
- Physical Barriers

Operational Rules:
- Safe Work Procedures
- RF Awareness Training

Note: Administrative controls are implemented only when engineering controls cannot sufficiently isolate the RF exposure.

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The Mandatory Mitigation Loop



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Final Deliverables and Certification



Report Components

- [x] Site Description & Diagrams
- [x] Antenna Inventory & Technical Parameters
- [x] RF Modelling Assumptions
- [x] Exposure Calculations & Cumulative Evaluation
- [x] Mitigation Recommendations (if any)
- [x] Final Compliance Statement

Analyst Responsibility Summary

The SC6 Analyst functions as the final checkpoint—validating inputs, evaluating cumulative risks across all operators, and ensuring every certified site protects public and occupational safety beneath the 100% threshold.

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